

Incorporating Sparse Arrays into a Cell-Free Near-Field / Far-Field XL-MIMO Uplink

Motivation, theory, design choices, references, and simulation evidence

Scope. This note explains why a sparse per-access-point (AP) array is worth adding to the existing cell-free hybrid near-field/far-field XL-MIMO uplink, what concrete benefits it brings, where the idea originates, which references support each design choice, how the advantages can be exploited, and what the five selected figures produced by the simulation code demonstrate. Closed-form (analytic) results are stated exactly; quantities that depend on the Monte-Carlo run are labelled as such so nothing is overstated.

1. Why add a sparse array?

The system places $N = 64$ antennas per AP at half-wavelength spacing ($d = \lambda/2$) at $f_c = 3$ GHz ($\lambda = 0.1$ m). The per-AP aperture is therefore $D = (N-1) \cdot \lambda/2 = 3.15$ m, and the Rayleigh (near-field) distance is

$$d_{\text{Ray}} = 2 D^2 / \lambda = 2 \cdot (3.15)^2 / 0.1 \sim 198 \text{ m}.$$

Two hard constraints motivate a sparse array:

- **The half-wavelength array cannot be made smaller.** Reducing N shrinks D , which shrinks d_{Ray} (quadratically), so the prominent near-field region on which the proposed Cross-AP list-SIC detector relies would disappear. At $\lambda/2$ spacing, N is effectively floored.
- **Many antennas are expensive.** Each element needs an RF chain and contributes to fronthaul and detector complexity. For XL-MIMO this is the dominant hardware cost.

A **sparse array** increases the element spacing to $d = s \cdot \lambda/2$ (sparsening factor $s > 1$). Because d_{Ray} depends on the *aperture*, not the element count, the same near-field distance is reached with far fewer antennas, or a much larger near-field region is obtained at the same N . The price is grating lobes, which the near-field regime makes range-selective rather than eliminating (Section 4). The remainder of this note quantifies this trade.

2. Where the idea comes from

Sparse arrays are a classical antenna concept (thinned/minimum-redundancy arrays, nested and coprime arrays) recently revisited for near-field XL-MIMO and ‘sparse MIMO’ communications. The specific results used here come from:

- **Sparse MIMO paradigm & interference suppression** for near- and far-field communications [R5], which shows sparse spacing raises the spatial degrees of freedom and improves interference rejection when users are in the near field.
- **Near-field beam-focusing pattern and grating-lobe characterization for modular XL-arrays** [R1], the origin of the 2-D (angle x range) beam-focusing analysis and the statement that the spherical wavefront makes grating lobes range-selective.
- **Advantages of sparse arrays in near-field XL-MIMO (EDoF)** [R2]: a closed form showing the effective degrees of freedom increase with sparsity up to a bound.
- **Near-field XL-MIMO tutorial** [R3] for the non-uniform spherical-wave (NUSW) model and the aperture / Rayleigh-distance relationship.
- **Reconfigurable array thinning** [R4] and **quasi-distributed (ULA to MRA) grating-lobe** analysis [R6], which frame the uniform-vs-non-uniform choice and the distributed setting.

The link to the group's own line of work is the detector: the near-field spherical wavefront is exactly what lets the Cross-AP list-SIC / IDD receiver separate users that a far-field array cannot, so anything that enlarges the near-field region feeds that receiver more of the structure it exploits.

3. System model and mathematical formulation

Array response. Let p_m be the position of element m along the array axis (centred). Under the plane-wave (far-field) model the steering vector depends on angle only:

$$a_{FF,m}(\theta) = \exp(j 2 \pi p_m \sin(\theta) / \lambda).$$

Under the non-uniform spherical-wave (NUSW) near-field model it depends on angle *and* range r through the exact element-to-source distance:

$$d_m(r, \theta) = \sqrt{ r^2 + p_m^2 - 2 r p_m \sin(\theta) },$$

$$a_{NF,m}(r, \theta) = (r / d_m) \cdot \exp(- j 2 \pi d_m / \lambda).$$

Grating lobes. For uniform spacing d , a beam steered to θ_0 has ambiguous replicas at

$$\sin(\theta_g) = \sin(\theta_0) \pm m \cdot \lambda / d, \quad m = 1, 2, \dots$$

that are visible whenever $|\sin(\theta_g)| \leq 1$. With $d = s \cdot \lambda / 2$ this becomes $\sin(\theta_g) = \sin(\theta_0) \pm 2m/s$, so for $s = 4$ and broadside steering the grating lobes sit at $\sin(\theta_g) = \pm 0.5$ and ± 1.0 (i.e. $\pm 30^\circ$ and $\pm 90^\circ$). In the far field these replicas reach the full main-lobe height (0 dB): a genuine ambiguity. In the near field the non-linear phase in $d_m(r, \theta)$ prevents them from adding up coherently except near the focal range, so they become range-selective [R1].

Rayleigh distance and antenna count. With aperture $D = (N-1) \cdot s \cdot \lambda / 2$,

$$d_{Ray}(s) = 2 D^2 / \lambda = (N-1)^2 s^2 \lambda / 2 \Rightarrow d_{Ray}(s) = s^2 \cdot d_{Ray}(1).$$

To hold a target near-field distance d_{Ray} while sparsening, the required antenna count is

$$N(s) = 1 + \sqrt{ 2 d_{Ray} / \lambda } / s.$$

For $d_{Ray} = 198$ m and $\lambda = 0.1$ m this gives $N(1) = 64$ (the baseline) and $N(4) \sim 17$, i.e. a **74% reduction in antennas (and RF chains) for the same near-field coverage** — a closed-form, assumption-free result.

Collinear users. Two users at the same angle θ but different ranges r_1, r_2 have identical far-field steering, $a_{FF}(\theta)$, so the 2-user channel is rank-deficient and no linear receiver can separate them: the post-MMSE SINR saturates (0 dB ceiling) at any SNR. Their near-field signatures $a_{NF}(r_1, \theta)$, $a_{NF}(r_2, \theta)$ differ, and a larger (sparse) aperture makes them more orthogonal, so they become separable [R5]. This is the mechanism the Cross-AP detector uses.

4. Benefits and how to exploit them

The advantages of a sparse array can be exploited along four axes; each is tied to a reference and to a concrete way of using it in this system.

#	Benefit	Mechanism / how to exploit	Ref
B1	Antenna & RF-chain reduction	Same d_{Ray} with ~74% fewer elements ($N(s)=1+\sqrt{2 d_{\text{Ray}}/\lambda/2}$). Use to cut per-AP hardware.	[R1,R2]
B2	Larger near-field region	$d_{\text{Ray}}(s)=s^2 d_{\text{Ray}}$. At fixed N, sparsening pushes more user-AP links into the near field, where the	[R3,R5]
B3	Range resolution / EDoF	Larger aperture raises effective DoF and resolves collinear (same angle) users by range that the far	[R2,R5]
B4	Grating lobes made range-resolved	Near-field spherical wavefront defocuses replicas away from the plane range; cross-AP fusion further	[R6]

4.1 Uniform vs non-uniform sparse array

Recommendation: start with a uniform sparse ULA; keep non-uniform (thinned / nested / coprime / minimum-redundancy) as a robustness comparator, and let the BER after cross-AP combining decide.

Reasoning (reference-backed, not the manifold argument). A uniform sparse array places its grating lobes at *predictable* angles ($\sin = 2m/s$) and is exactly what the modular/sparse XL-MIMO literature models [R1,R3], so results are directly comparable to that body of work and the grating angles are analytically tractable. A non-uniform array trades the discrete grating spikes for a raised sidelobe floor everywhere — better worst-case robustness to arbitrary interferer geometry, at the cost of a higher average sidelobe and an irregular manifold [R6, nested/coprime arrays].

Correction to a common justification. The irregular-manifold penalty of non-uniform arrays matters only for codebook / beam-training / polar-domain processing. The present detector builds MMSE combiners from *estimated* channels and uses no angle-range dictionary, so the 'preserve the structured manifold' argument does not apply here. The correct reasons to prefer uniform are comparability and analytical tractability — and, if a near-field channel-estimation codebook is added later, the manifold argument then becomes valid future-work justification.

5. What the selected figures show

The code produces five curated figures (set `sparse_study = true`). A sixth, the 2-D near-field focusing pattern, is available behind `show_mechanism = true` as an optional mechanism illustration.

Fig. S1 — Antenna / complexity reduction at equal near-field distance

What it shows. d_{Ray} vs N for $\lambda/2$ vs sparse spacing, and the antenna count required to hold the baseline d_{Ray} as a function of the sparsening factor s .

Result. Closed-form: holding $d_{\text{Ray}} = 198$ m needs $N \sim 17$ at $s = 4$ versus $N = 64$ at $\lambda/2$, a 74% reduction in antennas and RF chains. This is the strongest, assumption-free argument for the array.

Fig. S2 — Near-field link fraction vs sparsening (this deployment)

What it shows. A Monte-Carlo over the actual 300 x 300 m deployment measuring the percentage of user-AP links that are near-field (link distance $< d_{\text{Ray}}(s)$) as s grows.

Result. Simulation (read the printed values): the near-field fraction increases monotonically with s because $d_{\text{Ray}}(s)=s^2 d_{\text{Ray}}$; the $s=1$ and $s=4$ percentages are printed to the console. This is the evidence that sparse arrays are *relevant* to this geometry rather than assumed to be.

Fig. S3 — Collinear-user SINR, far/near x dense/sparse

What it shows. Post-MMSE SINR of a desired user against one collinear interferer (same angle, $\Delta r = 2$ m) for the four combinations of {far-field, near-field} x {dense, sparse}.

Result. Analytic + simulation: both far-field curves saturate at the 0 dB rank-deficiency ceiling (identical steering, unrecoverable at any SNR); the near-field curves rise with SNR, and the near-field sparse array rises

fastest because its larger aperture gives the finest range resolution. This is the detection-relevant benefit that connects to the Cross-AP receiver.

Fig. S4 — Grating-lobe cost and its near-field mitigation

What it shows. Left: the beam gain vs angle at $s = 4$ for the sparse array under the far-field model (full-height grating lobes) vs the near-field model (range-selective), with a dense reference. Right: the peak grating-lobe level vs element spacing, far-field vs near-field.

Result. Simulation (from your run): under the far-field model the grating replicas reach ~ 0 dB (full ambiguity); under the near-field model they are held roughly 6-9 dB lower. The honest reading is that the near field provides a ~ 6 -9 dB range-selective mitigation, NOT immunity — a residual remains at the focal range. State it as 'range-selective', never 'removed'.

Fig. S5 — Cross-AP combining vs the per-AP grating lobe (novel test)

What it shows. An interferer is placed exactly on AP#1's grating angle ($\sin = 2/s$), where AP#1's sparse array cannot null it; the other APs see it at ordinary angles. Desired-user SINR is compared at AP#1 alone vs after cross-AP LSFD fusion, dense vs sparse.

Result. Simulation (read the printed dB recovery): the sparse AP#1-alone curve sits below the dense one (the grating-lobe liability is real); cross-AP fusion is expected to recover most of that loss because the remaining APs resolve the interferer. The printed 'recovers X dB' line is the number that turns 'sparse arrays have interesting beampatterns' into 'sparse arrays work in this detector'. If fusion does NOT recover it, that is a legitimate negative result to report, not a failure.

6. Honest limitations and the open question

- **The grating lobe is a real cost.** The near field mitigates it (~ 6 -9 dB, range-selective); it does not remove it. Avoid the words 'suppressed' / 'immune' in the paper.
- **Benefit is conditional on the near-field fraction.** Sparse helps only for links that are in the near field; far-field APs gain nothing and still see fully-ambiguous grating lobes. Fig. S2 quantifies how many links are near-field, which is why it must accompany the benefit figures.
- **Figs. S1-S5 are per-array / per-scene analyses.** The genuinely novel, still-open claim — that cell-free cross-AP combining suppresses the residual grating lobes across APs — is only probed at the single-interferer level in Fig. S5. The decisive evidence is a full system-level BER / sum-rate comparison of {dense $\lambda/2$, uniform sparse} under the four detectors, which requires threading the sparse geometry into the main Monte-Carlo (channel generation, near-field mask). That is the recommended next step.

7. Conclusions and recommended next steps

The defensible claims, in order of strength:

- **(1) Hardware:** a uniform sparse array reaches the same near-field distance with $\sim 74\%$ fewer antennas ($N: 64 \rightarrow 17$ at $s = 4$) — closed-form, unconditional.
- **(2) Detection:** the resulting large aperture resolves collinear users by range that a far-field array fundamentally cannot, directly benefiting the Cross-AP list-SIC / IDD receiver.
- **(3) Relevance:** on the actual 300 m deployment, sparsening moves a measurable fraction of links into the near field (Fig. S2), which is the precondition for (2).
- **(4) Cost, honestly:** sparsity causes grating lobes; the near field makes them range-selective (~ 6 -9 dB), and cross-AP fusion is the mechanism proposed to handle the residual (Fig. S5).

Next step. Run Figs. S1-S5. If S2 shows a meaningful near-field-fraction increase and S5 shows cross-AP fusion recovering the grating-lobe loss, proceed to the system-level BER/sum-rate integration with the sparse per-AP geometry — that figure decides whether sparse arrays belong in the paper. If S5 shows fusion does not recover

the loss, report that as the finding and reconsider (e.g. non-uniform or geometry-aware detection) before investing in the full integration.

References

- [R1] H. Wang, Z. Xiao, Y. Zeng, *et al.*, “Near-Field Beam Focusing Pattern and Grating Lobe Characterization for Modular XL-Array,” arXiv:2305.05408, 2023.
- [R2] “Exploring the Advantages of Sparse Arrays in Near-Field XL-MIMO Systems: Beam Analysis and EDoF Function,” arXiv:2501.09234, 2025.
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- [R4] “Sparse Array Design for Near-Field MU-MIMO: A Reconfigurable Array Thinning Approach,” arXiv:2602.21973.
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- [R6] “Near-Field Communications with Grating Lobes for Quasi-Distributed Arrays: From ULA to MRA,” arXiv:2607.29341.
- [R7] P. Pal and P. P. Vaidyanathan, “Nested Arrays: A Novel Approach to Array Processing with Enhanced Degrees of Freedom,” IEEE Trans. Signal Process., 2010; P. P. Vaidyanathan and P. Pal, “Sparse Sensing with Coprime Arrays,” 2011 (non-uniform array DoF).
- [R8] M. Cui and L. Dai, “Channel Estimation for Extremely Large-Scale MIMO: Far-Field or Near-Field?,” IEEE Trans. Commun., 2022 (polar-domain / near-field manifold).

Prepared as supporting material for the sparse-array extension of the cell-free NF/FF XL-MIMO uplink code. Analytic results are exact; simulation-dependent quantities should be filled from the console output of your own run.